

BUSINESS REQUIREMENTS DOCUMENT

Product Description:

Expense Tracker is an API based expense management tool that helps is easy storage, monitoring, retrieval and reporting summary according to categories such as shop and time(weekly, monthly, yearly). Users can enter their expenses and do not need to worry about tracking & calculating expenses, which is automatically done by the product without needing tedious manual work.

Users benefited:

- People who want to store, track and report day-to-day expenses.
- Small-scale organizations that need expense maintenance & tracking without much manual intervention.
- Individuals who lack math and require assistance in expense tracking & summaries.
- Busy professionals who don't want to spend much time tracking & calculating every individual expense.
- Anyone who wants to understand his/her spending characteristics and regulate their expenses based on the data.

The 'Why?':

- Easy to understand
- Easy maintenance
- Doesn't require technical knowledge/prerequisites
- Easy installation & setup
- No messy and incorrect expense storage all over the place
- Single secured storage
- Auto-formatted data while storing
- No manual calculations
- Quick data retrieval and reporting

Assumptions:

- All amounts entered are in single currency & no multi-currency conversion required.
- PostgreSQL is installed and running locally.
- The value entered by the customer is expected to be correct. The data being entered includes date, description and amount (i.e., the entered date is expected to be true as entered by the customer without needing to verify whether the expenditure was made on that date).
- No time zone handling.
- The user has minimal computer literacy.

Risks & Mitigations:

- No secured accounts for every user to use the portal. This will be made personalized and secure in the near future.
- Database connection failure in the start can lead to a lack of data storage & errors. Connections will be made easy and cloud-based without needing to install & set up PgAdmin locally.
- Multiple users on a session can cause session conflicts. This will be adjusted when creating personalized account handling.

Glossary:

- a) API - Application Programming Interface — a set of rules allowing systems to communicate
- b) REST - Representational State Transfer an architectural style for APIs using HTTP
- c) CRUD - Create, Read, Update, Delete the four basic database operations
- d) ORM - Object-Relational Mapper maps Python classes to database tables
- e) FastAPI - A modern Python web framework for building APIs with automatic documentation
- f) SQLAlchemy - Python SQL toolkit and ORM library
- g) Pydantic - Python data validation library used for request/response schemas
- h) PostgreSQL - Open-source relational database management system
- i) SQLite - Lightweight file-based database used for testing
- j) Swagger UI - Auto-generated interactive API documentation interface
- k) pytest - Python testing framework
- l) BRD - Business Requirements Document describes what the system must do from a business perspective
- m) Endpoint - A specific URL in the API that handles a particular type of request
- n) HTTP Status Code - A 3-digit code indicating the result of an HTTP request (200=OK, 404=Not Found, etc.)
- o) Dependency Injection - A design pattern where dependencies are passed in rather than created internally
- p) Fixture - A reusable setup function in pytest that provides test dependencies
- q) Rollback - Undoing database changes used in tests to reset state after each test
- r) Environment Variable - A variable set outside the code, used to store configuration like passwords
- s) Schema - Defines the structure and validation rules for data
- t) Migration - A versioned change to the database schema